



SLIDES BY

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Chapter 6, Part B

Time Series Analysis and Forecasting

- Linear Trend Projection
- Seasonality

Linear Trend Projection

- If a time series exhibits a linear trend, the method of **least squares** may be used to determine a trend line (projection) for future forecasts.
- Least squares, also used in regression analysis, determines the unique **trend line forecast** which **minimizes** the **mean square error** between the trend line forecasts and the actual observed values for the time series.
- The **independent variable** is the **time period** and the **dependent variable** is the **actual observed value in the time series**.

Linear Trend Projection

- Using the method of least squares, the formula for the trend projection is:

$$T_t = b_0 + b_1 t$$

where: T_t = linear trend forecast in period t
 b_0 = Y-intercept of the linear trend line
 b_1 = slope of the linear trend line
 t = the time period

Linear Trend Projection

- For the trend projection equation $T_t = b_0 + b_1 t$

$$b_1 = \frac{\sum_{t=1}^n (t - \bar{t})(Y_t - \bar{Y})}{\sum_{t=1}^n (t - \bar{t})^2}$$

$$b_0 = \bar{Y} - b_1 \bar{t}$$

where: Y_t = actual value of the time series in period t

n = number of periods in the time series

$\bar{Y}$ = average values of the time series

$\bar{t}$ = mean value of t

Linear Trend Projection

■ Example: Auger's Plumbing Service

The number of plumbing repair jobs performed by Auger's Plumbing Service in the last nine months is listed on the right.

Forecast the number of repair jobs Auger's will perform in December using the least squares method.

<u>Month</u>	<u>Jobs</u>	<u>Month</u>	<u>Jobs</u>
March	353	August	409
April	387	September	399
May	342	October	412
June	374	November	408
July	396		

Linear Trend Projection

■ Example: Auger's Plumbing Service

(month) t	$t - \bar{t}$	$(t - \bar{t})^2$	Y_t	$(Y_t - \bar{Y})$	$(t - \bar{t})(Y_t - \bar{Y})$
(Mar.) 1	-4	16	353	-33.67	134.68
(Apr.) 2	-3	9	387	0.33	-0.99
(May) 3	-2	4	342	-44.67	89.34
(June) 4	-1	1	374	-12.67	12.67
(July) 5	0	0	396	9.33	0
(Aug.) 6	1	1	409	22.33	22.33
(Sep.) 7	2	4	399	12.33	24.66
(Oct.) 8	3	9	412	25.33	75.99
(Nov.) 9	4	16	408	21.33	85.32
Sum 45	$t_{\text{ngang}} = 560$		3480		444.00

$t_{\text{ngang}} = 5$

Y_{ngang}

Linear Trend Projection

■ Example: Auger's Plumbing Service

$$\bar{t} = 45 / 9 = 5 \quad \bar{Y} = 3480 / 9 = 386.667$$

$$b_1 = \frac{\sum_{t=1}^n (t - \bar{t})(Y_t - \bar{Y})}{\sum_{t=1}^n (t - \bar{t})^2} = \frac{3480}{60} = 7.12$$

$Y_t = 7.12 * t + 351.07$

$$b_0 = \bar{Y} - b_1 \bar{t} = 386.667 - 7.12(5) = 351.07$$

$$T_{10} = 351.07 + (7.12)(10) = 422.27$$

Linear Trend Projection

■ Example: Auger's Plumbing Service

Forecast for December (Month 10) using a three-period ($k = 3$) weighted moving average with weights of .6, .3, and .1 for the newest to oldest data, respectively. Then, compare this Month 10 weighted moving average forecast with the Month 10 trend projection forecast.

<u>Month</u>	<u>Jobs</u>	<u>Month</u>	<u>Jobs</u>
March	353	August	409
April	387	Septem.	399
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Linear Trend Projection

■ Example: Auger's Plumbing Service

• Three-Month Weighted Moving Average

The forecast for December will be the weighted average of the preceding three months: September, October, and November.

$$\begin{aligned} F_{10} &= .1Y_{\text{Sep.}} + .3Y_{\text{Oct.}} + .6Y_{\text{Nov.}} \\ &= .1(399) + .3(412) + .6(408) \\ &= \underline{\underline{408.3}} \end{aligned}$$

• Trend Projection

$$F_{10} = \underline{\underline{422.27}} \text{ (from earlier slide)}$$

Linear Trend Projection

- Example: Auger's Plumbing Service
 - Conclusion

Due to the positive trend component in the time series, the trend projection produced a forecast that is more in line with the trend that exists. The weighted moving average, even with heavy (.6) weight placed on the current period, produced a forecast that is lagging behind the changing data.

Seasonality without Trend

- To the extent that seasonality exists, we need to incorporate it into our forecasting models to ensure accurate forecasts.
- We will first look at the case of a seasonal time series with no trend and then discuss how to model seasonality with trend.

Seasonality without Trend

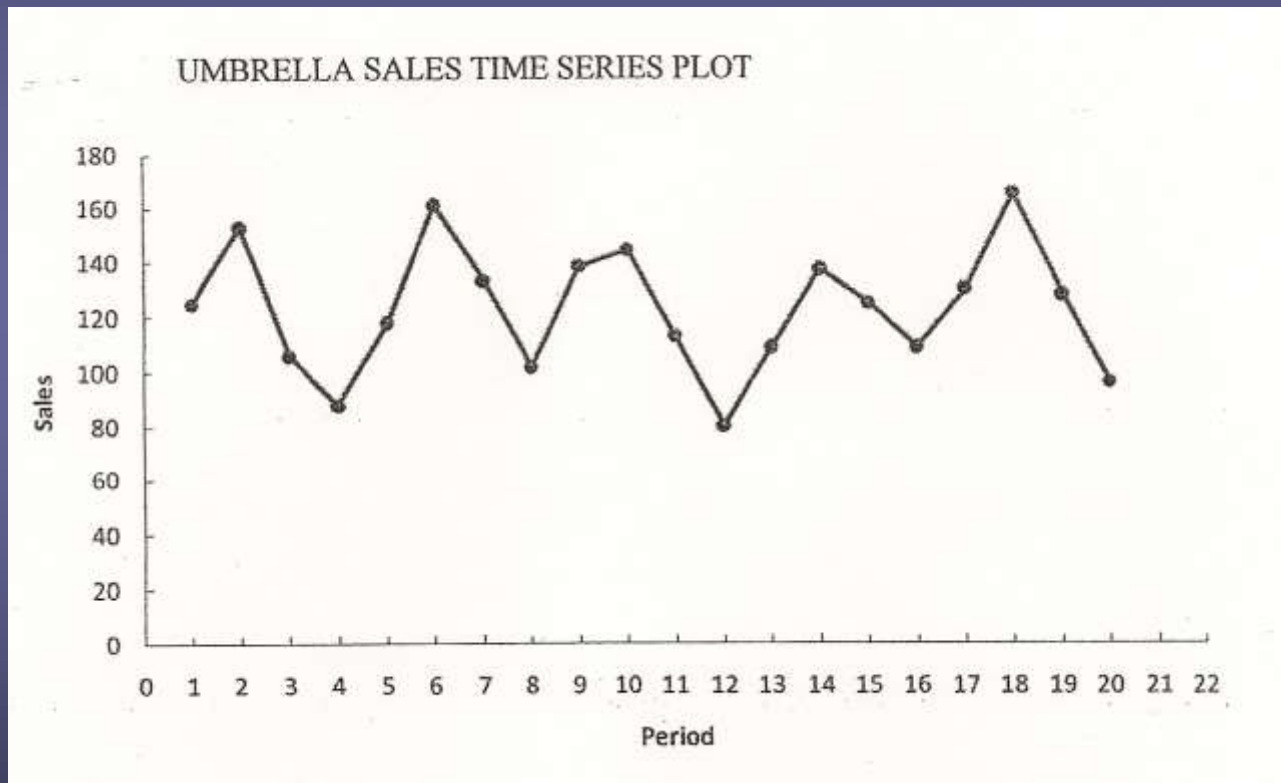
- Example: Umbrella Sales

<u>Year</u>	<u>Quarter 1</u>	<u>Quarter 2</u>	<u>Quarter 3</u>	<u>Quarter 4</u>
1	125	153	106	88
2	118	161	133	102
3	138	144	113	80
4	109	137	125	109
5	130	165	128	96

- Sometimes it is difficult to identify patterns in a time series presented in a table.
- Plotting the time series can be very informative.

Seasonality without Trend

- Example: Umbrella Sales
 - Time Series Plot



Seasonality without Trend

■ Example: Umbrella Sales

- The time series plot does not indicate any long-term trend in sales.
- However, close inspection of the plot does reveal a seasonal pattern.
 - The first and third quarters have moderate sales,
 - the second quarter the highest sales, and
 - the fourth quarter tends to be the lowest quarter in terms of sales.

Seasonality without Trend

- We will treat the season as a categorical variable.
- Recall that when a categorical variable has k levels, $k - 1$ **dummy variables** are required.
- If there are four seasons, we need three dummy variables.
 - $Qtr1 = 1$ if Quarter 1, 0 otherwise
 - $Qtr2 = 1$ if Quarter 2, 0 otherwise
 - $Qtr3 = 1$ if Quarter 3, 0 otherwise

Seasonality without Trend

■ Example: Umbrella Sales

- General Form of the Equation is:

$$F_t = b_0 + b_1(Qtr1_t) + b_2(Qtr2_t) + b_3(Qtr3_t)$$

- Optimal Model is:

$$Sales_t = 95.0 + 29.0(Qtr1_t) + 57.0(Qtr2_t) + 26.0(Qtr3_t)$$

- The forecasts of quarterly sales in year 6 are:

- Quarter 1: Sales = $95 + 29(1) + 57(0) + 26(0) = 124$
- Quarter 2: Sales = $95 + 29(0) + 57(1) + 26(0) = 152$
- Quarter 3: Sales = $95 + 29(0) + 57(0) + 26(1) = 121$
- Quarter 4: Sales = $95 + 29(0) + 57(0) + 26(0) = 95$

Seasonality without Trend

■ Example: Umbrella Sales

<u>Year</u>	<u>Quarter 1</u>	<u>Quarter 2</u>	<u>Quarter 3</u>	<u>Quarter 4</u>
1	125	153	106	88
2	118	161	133	102
3	138	144	113	80
4	109	137	125	109
5	130	165	128	96
Average	124	152	121	95

- We could have obtained the quarterly forecasts for next year by simply computing the average number of umbrellas sold in each quarter.

Seasonality and Trend

- We will now extend the curve-fitting approach to include situations where the time series contains both a seasonal effect and a linear trend.
- We will introduce an additional variable to represent time.

Seasonality and Trend

■ Example: Terry's Tie Shop

Business at Terry's Tie Shop can be viewed as falling into three distinct seasons: (1) Christmas (November and December); (2) Father's Day (late May to mid June); and (3) all other times.

Average weekly sales (\$) during each of the three seasons during the past four years are shown on the next slide.

Seasonality and Trend

- Example: Terry's Tie Shop

<u>Year</u>	<u>Season</u>		
	<u>1</u>	<u>2</u>	<u>3</u>
1	1856	2012	985
2	1995	2168	1072
3	2241	2306	1105
4	2280	2408	1120

Determine a forecast for the average weekly sales in year 5 for each of the three seasons.

Seasonality and Trend

- Example: Terry's Tie Shop

- There are three seasons, so we will need two dummy variables.

- $Seas1_t = 1$ if Season 1 in time period t , 0 otherwise
 - $Seas2_t = 1$ if Season 2 in time period t , 0 otherwise

- General Form of the Equation is:

$$F_t = b_0 + b_1(Seas1_t) + b_2(Seas2_t) + b_3(t)$$

- Optimal Model

$$Sales_t = 797.0 + 1095.43(Seas1_t) + 1189.47(Seas2_t) + 36.47(t)$$

Seasonality and Trend

- Example: Terry's Tie Shop

- The forecasts of average weekly sales in the three seasons of year 5 (time periods 13, 14, and 15) are:

$$\begin{aligned}\text{Seas. 1: Sales}_{13} &= 797 + 1095.43(1) + 1189.47(0) + 36.47(13) \\ &= 2366.5\end{aligned}$$

$$\begin{aligned}\text{Seas. 2: Sales}_{14} &= 797 + 1095.43(0) + 1189.47(1) + 36.47(14) \\ &= 2497.0\end{aligned}$$

$$\begin{aligned}\text{Seas. 3: Sales}_{15} &= 797 + 1095.43(0) + 1189.47(0) + 36.47(15) \\ &= 1344.0\end{aligned}$$

End of Chapter 6, Part B

